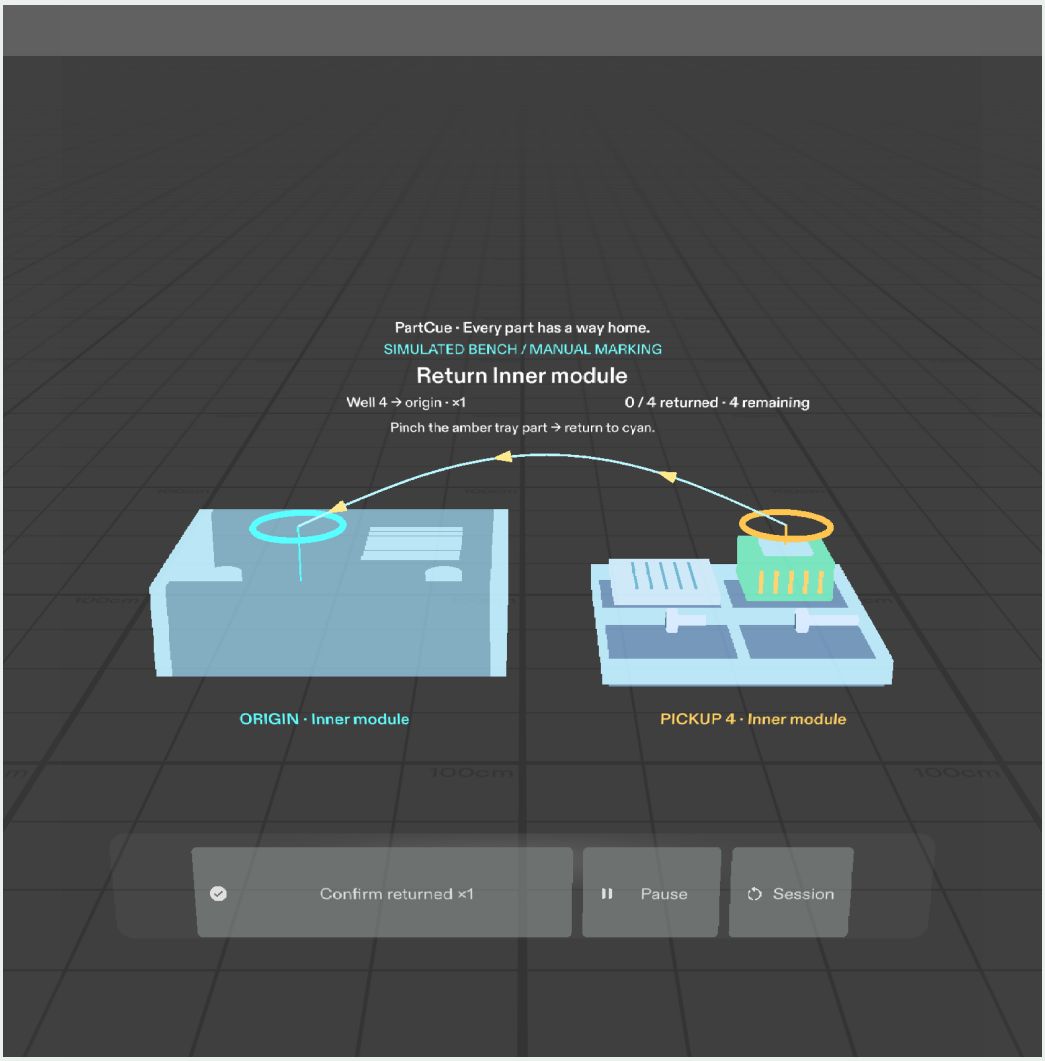
A SPATIAL MEMORY FOR THE REPAIR BENCH

PartCue

Every part has a way home.

Two similar screws remain. Which one came from here?

ACTUAL LENS STUDIO PREVIEW / SIMULATED BENCH / MANUAL MARKING



Record a part, then follow the paired tray-to-origin cue when returning it.

CREATOR / PRODUCT OWNER

Shivam Gupta

AI-assisted implementation

THE CORE INTERACTION

Remember the way back.

A record of what the user actually removed, with an explicit confirmation for each return.

1. **Mark the removal**

   In the Lens, select an authored part and tray well. In a custom browser job, write the part name, origin, container, quantity and optional note.

2. **Follow one pair**

   Return mode reverses the confirmed removal log. Select the highlighted tray target; a directional path connects pickup to the saved origin.

3. **Confirm, correct, continue**

   A wrong selection preserves progress. One active confirmation stays beside the task. Phase-specific controls recede; pause, resume, and undo preserve context.

4. **Finish with a reconciled log**

   Completion means every recorded group has a user-confirmed return. It does not certify assembly, safety, or device function.

Why glasses?

The useful information is a relationship between two places on the bench. Keeping that relationship beside the work could reduce looking away to reconstruct photographs or notes.

SCOPE / SIMULATED BENCH AND MANUAL MARKING

The Lens uses **four authored parts and predefined origins**. Its session is separate from the browser. The hosted companion accepts up to 100 custom part groups with written references and an illustrative diagram. Private Firebase accounts save browser jobs; JSON files support deliberate backups and handoffs.

A PRODUCT MUST EARN ITS PLACE

Start at the repair bench.

Independent repair shops and repair classrooms are the proposed first customers. Occasional home repair is a secondary free-use segment.

$20 **/ active bench / month**

Unvalidated price experiment. Bring your own supported glasses.
A free single-session mode would let users evaluate the workflow.

ILLUSTRATIVE MONTHLY CONTRIBUTION / EVERY FIGURE IS AN ASSUMPTION

Subscription revenue	**$20.00**
Payment fee: assumed 3% + $0.30	**- $0.90**
Optional text storage and transfer	**- $0.50**
Support: 10 minutes at $12/hour	**- $2.00**
Contribution before other costs	**$16.60**

This is illustrative arithmetic, not a margin forecast. It excludes hardware, development, acquisition, sales, taxes, refunds, fleet management, insurance, and other business costs. Early support cost could be much higher.

CUSTOMER-SIDE TEST

60 net minutes saved each month would cover the software price at an assumed $20/hour labor value. That excludes glasses and onboarding. Net savings must count time spent marking and correcting the log.

THE HYPOTHESIS CAN FAIL

If recording overhead exceeds recall savings, or users prefer labeled mats and bags, simplify the capture flow or stop this wedge. No customer demand or time saving has been established.

A NARROW ADVANTAGE TO PROVE

The reverse log already exists.

PartCue must earn its place through spatial pickup-to-origin cues, not claim invention of teardown journals or reverse checklists.

01 Direct reverse-checklist baseline

KanadeK/parttrail describes local teardown capture, quantities, tray locations, portable JSON, and reverse reassembly. The README is primary evidence; its runtime has not been tested here.

02 iFixit mats and trays

Writable mats already preserve screw locations cheaply. Observed US prices: $19.95 for Magnetic Project Mat; $36.95 for FixMat. A subscription must justify itself.

03 PTC Vuforia / TeamViewer Frontline

Enterprise AR guidance, capture, assembly, and inspection workflows are established. Our specific paired spatial cue is a hypothesis to validate, not an exclusive-feature claim.

PILOT PROPOSAL / NOT YET CONDUCTED

Five technicians. Matched battery-free fixtures. Counterbalanced trials with usual methods, a plain reverse checklist, and PartCue spatial mode. Count all recording and correction time.

01 No unreported origin or quantity mismatch in the final fixture check.

02 At least 10% lower median total time than usual methods; spatial mode must also beat the plain reverse checklist without reducing accuracy.

03 Four of five complete a second session unaided and prefer another use.

04 Three of five accept a $20/bench/month paid pilot offer.

A small pilot gives directional evidence. There is no established moat. Better alignment recovery, workflow fit, and earned trust remain possible future advantages.

PRIMARY REFERENCES / RESEARCH SNAPSHOT: 20 OCTOBER 2026

KanadeK/parttrail: primary README
iFixit Magnetic Project Mat
iFixit FixMat
PTC Vuforia: AR work instructions
TeamViewer Frontline

EVIDENCE SNAPSHOT / 20 OCTOBER 2026

Built, tested, ready to demonstrate.

A working Lens, a hosted job companion, captured workflows and public source. Software, cloud and Lens results are reported separately.

51	12	7/7
SOFTWARE TESTS	FIREBASE TESTS	LENS SCENARIOS

Automated software checks and manual browser QA are separate from Lens validation.

OBSERVED IN THE COMPANION BROWSER FLOW

Practice and custom named-part jobs; wrong-part and quantity rejection; pause, resume, undo and completion. Real hosted registration, save/reload/load, private account isolation, concurrent revision rejection, valid import with a new ID, and invalid import preserving current work.

LENS IMPLEMENTATION AND VERIFICATION

Built through CLAD and official Lens Studio MCP. Seven named functional and IK scenarios passed in preview after the latest stale-selection guard. This count excludes the demonstration choreography. The Lens core exactly matches the shared TypeScript source. Real tray pickup and Confirm use far-field routing.

WHAT REMAINS UNPROVEN

Four authored parts and predefined origins define the preview scope. Marks and return confirmations are manual. Lens state is session-only, with no automatic companion sync. Camera recognition and physical installation verification are absent. Device tracking, comfort, technician time savings, and willingness to pay need separate studies.

Release status

DONE	Hosted at partcue.web.app; public source, planning and repeatable checks.
DONE	Captured the workflow, spatial-viewpoint clip, and final runtime views.
DONE	Published the captioned public product demonstration (2:37).
NEXT	Complete personal details and eligibility, then select final Submit on Devpost.

Next: a measured pilot at the repair bench.